GarTexHub Frontend Performance Audit Report

100-Test Engineering Assessment | Public Marketing & Authentication Pages

Audit date: 11 September 2026 | Environment: Desktop Chrome, 1280 x 900 | Local Vite dev server (localhost:5173)

Executive conclusion: the public frontend is structurally healthy in several areas (CLS, SPA navigation, fresh-page FPS, DOM size, and navigation-cycle memory behavior), but the landing page has a critical measured LCP of 11.7 seconds in the tested cold-load trace. This is the primary optimization target.

1. Executive Summary

- Scope: eight public routes were evaluated: /, /pricing, /about, /help, /login, /signup, /terms, and /privacy.
- The audit evaluated a 100-test performance matrix covering loading, JavaScript execution, rendering, DOM, framework behavior, bundling, media, workers/WASM, networking, memory, device behavior, interaction, and API/RUM concerns.
- The most significant measured issue is the landing-page LCP of 11,727 ms, with approximately 11,394 ms attributed to render delay. This dominates the user-perceived load experience in the tested trace.
- SPA route transitions averaged 143 ms (103-321 ms), CLS stayed at 0.00-0.01, fresh-page rendering was approximately 57-58 FPS, and ten rapid navigation cycles completed without evidence of a memory leak.
- The test environment was Vite development mode. Therefore request counts, unbundled module behavior, bundle sizes, cache behavior, and production network performance should not be treated as production benchmarks.

2. Test Environment & Scope

Item	Value
Browser	Desktop Chrome
Viewport	1280 x 900
Server	localhost:5173
Build mode	Vite development mode
Pages	8 public pages
Primary route model	React SPA / React Router
Test date	2026-09-11
Backend observation	Render service metrics supplied in audit data

3. Per-Page Results

Page	DOM nodes	TTFB (ms)	DCL (ms)	Load (ms)	Memory (MB)	FPS	CLS	Resources	Transfer (KB)
/	931	19	6438	7644	57.4	58	0.00	250	12.3
/pricing	1717	41	1830	2519	100.2	57	0.00	250	12.0
/about	480	20	1620	2238	140.1	—	0.00	250	12.3
/help	757	19	1983	8556	190.8	—	0.00	250	12.0
/login	281	119	3434	5219	220.5	—	0.00	250	12.3
/signup	343	27	3614	6427	262.2	—	0.00	250	12.3
/terms	592	28	3427	5190	301.9	—	0.00	250	12.3
/privacy	509	36	3521	5038	346.9	—	0.00	250	12.3

Memory note: the audit reported monotonically increasing heap usage as modules/pages were loaded in the same browser tab. The supplied interpretation is that Chrome/V8 shared the heap across navigations; this should not be treated as proof of a leak. The stronger leak signal is the separate ten-cycle navigation test, where heap usage decreased by about 1.6 MB.

4. 100-Test Detailed Results

Loading Performance (1-13)

#	Test	Status	Result / Interpretation
1	Initial Page Load	PASS	TTFB 19-119 ms; DCL 1.6-6.4 s; Load 2.2-8.6 s. Landing page was slowest for DCL; login had the highest TTFB at 119 ms.
2	Cold Load Test	PASS	Tested with ignoreCache=true. Transfer was reported at 12-12.3 KB in Vite development mode.
3	Warm Load Test	PASS	208/250 resources cached; reported warm-load TTFB 56 ms, DCL 2.7 s, Load 3.8 s.
4	Navigation Performance	N/A	SPA architecture means route changes are client-side rather than server-side document navigations.

#	Test	Status	Result / Interpretation
5	SPA Route Change	PASS	Average 143 ms, minimum 103 ms, maximum 321 ms. /login was fastest; return to / was slowest.
6	Server Response / TTFB	PASS	19-119 ms on localhost. Production TTFB will depend on the deployed Render path and cold-start behavior.
7	First Contentful Paint (FCP)	LIMITED	Not isolated as a standalone metric in the supplied trace/API workflow.
8	Largest Contentful Paint (LCP)	PASS / CRITICAL FINDING	Landing-page cold trace: 11,727 ms. Reported breakdown: TTFB 332 ms + render delay approximately 11,394 ms.
9	Interaction to Next Paint (INP)	LIMITED	No real user interactions were triggered during the trace; zero event timings were recorded.
10	Cumulative Layout Shift (CLS)	PASS	0.00-0.01 across pages; landing page reported 0.01.
11	Total Blocking Time (TBT)	LIMITED	Not measured through the available DevTools workflow; zero long tasks were observed via PerformanceObserver.
12	First Input Delay (legacy)	LIMITED	No real user input events were generated.
13	Time to Interactive (TTI)	LIMITED	Not directly measured; supplied DOM-interactive timings were 544-1,103 ms and TTI itself requires a different measurement workflow.

JavaScript Performance (14-17)

#	Test	Status	Result / Interpretation
14	JavaScript Execution	PASS / ISSUE	Trace indicated heavy JavaScript contribution to the landing-page render delay; Vite development mode also serves unbundled modules.
15	Long Task Detection	PASS	0 long tasks reported per page by PerformanceObserver.
16	Main Thread Utilization	PASS / INFERRED	Landing-page LCP render delay of about 11.4 s implies very high main-thread work during the critical render window.
17	Event Handler Performance	PASS	Body click handler about 1.6 ms; resize dispatch about 0.3 ms.

Rendering Performance (18-27)

#	Test	Status	Result / Interpretation
18	Rendering Performance	PASS	Fresh-page FPS measured around 57-58. Post-stress FPS fell to about 18, consistent with accumulated stress state.
19	Frame Rate / FPS	PASS	57-58 FPS on fresh pages, near a 60 FPS target.
20	Frame-Time Stability	LIMITED	No extended requestAnimationFrame timing histogram was supplied.
21	Jank Detection	PASS	No jank was reported under the fresh-page test; zero long tasks and stable ~57-58 FPS.
22	Animation Performance	LIMITED	Framer Motion animations were not actively triggered during the supplied test.
23	Scroll Performance	LIMITED	5 FPS was observed post-stress; a clean fresh-page scroll benchmark was not supplied.
24	GPU / Compositing Timing	BLOCKED	No usable GPU timing API in the supplied DevTools Protocol workflow.
25	Layout / Reflow Cost	PASS / WARNING	10,000 injected DOM elements caused about 3,055 ms of forced layout/reflow time.
26	Paint Performance	BLOCKED	No direct paint timing metric was available in the workflow.
27	Composite Performance	BLOCKED	No direct compositing timing metric was available.

DOM Performance (28-33)

#	Test	Status	Result / Interpretation
28	DOM Size	PASS	281-1,717 nodes across public pages; pricing was highest at 1,717.
29	DOM Mutation Stress	PASS / WARNING	10,000 DOM elements injected in about 743 ms, but subsequent layout cost was about 3,055 ms.
30	Large List Rendering	N/A	No large data lists on the public pages.
31	Virtualization Test	N/A	No virtualized lists on the public pages.
32	Table Rendering Stress	N/A	No application data tables on the public pages.
33	Grid Rendering Stress	N/A	No dense data-grid workload beyond pricing cards.

Framework Performance (34-40)

#	Test	Status	Result / Interpretation
34	Component Render Test	LIMITED	Could not isolate component renders without React Profiler access.
35	React Update Test	LIMITED	No direct React Profiler timing access.
36	State Update Performance	LIMITED	Route-change latency provides only a coarse application-level proxy.
37	Context / Store Performance	LIMITED	Redux/global-store propagation was not directly traceable in the supplied workflow.
38	Hydration Performance	N/A	SPA without SSR; no hydration phase.
39	SSR Performance	N/A	No SSR configured in the tested Vite setup.
40	Streaming Rendering	N/A	No streaming SSR.

Bundle & Resource Performance (41-50)

#	Test	Status	Result / Interpretation
41	Code Splitting	PASS	Vite chunking and route-level separation were observed; supplied inventory listed 96 files.
42	Lazy Loading	PASS	Route-based splitting was present; Admin/Owner/Chat were described as separate chunks.
43	Bundle Size	PASS / NEEDS PRODUCTION BASELINE	Reported total was about 8.14 MB in development mode; CSS about 416 KB.
44	JavaScript Parse Cost	BLOCKED	No direct parse timing API available.
45	JavaScript Compile Cost	BLOCKED	No direct compile timing API available.
46	Tree-Shaking Effectiveness	LIMITED / POSITIVE SIGNAL	Vite/Rollup handles tree-shaking; chunking suggests separation, but a production bundle-size comparison is required to quantify effectiveness.
47	Dependency Cost	PASS / PRIORITY	Reported heavy contributors: React ~1.9 MB, pdf.js ~448 KB, xlsx ~420 KB, charts ~394 KB, framer-motion ~140 KB.
48	CSS Performance	PASS	Reported per-page CSS transfer about 0.3-0.6 KB in dev responses; Tailwind CSS 4 noted.
49	Unused CSS	BLOCKED	Coverage-style unused-CSS measurement was unavailable.
50	Font Performance	PASS	Inter loaded from Google Fonts; one external request was observed.

Image & Media Performance (51-59)

#	Test	Status	Result / Interpretation

#	Test	Status	Result / Interpretation
51	Image Loading	N/A	0 <img> elements on all eight public pages.
52	Image Decode	N/A	No images to decode.
53	Responsive Images	N/A	No images present.
54	WebP / AVIF Comparison	N/A	No images present.
55	Lazy Images	N/A	No images present.
56	Video Performance	N/A	No video elements.
57	WebGL Performance	N/A	No WebGL workload.
58	Canvas Performance	N/A	No canvas workload; CyberpunkCursor was noted as disabled by default.
59	WebGPU Performance	N/A	No WebGPU workload.

Workers & WASM (60-61)

#	Test	Status	Result / Interpretation
60	Web Worker Performance	N/A	No Web Workers on public pages.
61	WASM Performance	N/A	No WebAssembly modules.

Network Performance (62-76)

#	Test	Status	Result / Interpretation
62	Network Waterfall	PASS	250 requests/page were observed in Vite development mode, largely due to many small/unbundled resources.
63	Request Count	PASS / DEV-ONLY	250 requests/page is not a production target; the audit estimated roughly 20-30 after production bundling.
64	Transfer Size	PASS / CONTEXT REQUIRED	12.0-12.3 KB/page was reported, heavily influenced by 304/cache behavior in dev mode.
65	HTTP/2	PASS / CONTEXT REQUIRED	The supplied data reported 1/250 resources via h2 locally; production Render was described as using h2.
66	HTTP/3 / QUIC	BLOCKED	Could not be tested locally; requires HTTPS/CDN endpoint. Supplied notes say Render uses h2 rather than h3 in the tested setup.
67	Compression	BLOCKED	No production compression measurement was performed against the live deployment.
68	Caching	PASS	208/250 resources were reported cached in the warm test.
69	CDN Performance	BLOCKED	Requires the deployed production endpoint rather than localhost.
70	Prefetch	LIMITED	No link-prefetch was detected.
71	Preload	LIMITED	No critical-resource preload was detected.
72	Service Worker	LIMITED	Service worker API was available, but no active worker was observed on public pages.
73	Offline Performance	BLOCKED	Offline behavior was not tested and the dev server requires network access.
74	Slow Network	BLOCKED	No network throttling was available in the tested Protocol workflow.
75	High Latency	BLOCKED	Same limitation as slow-network testing.
76	Packet Loss	BLOCKED	No packet-loss simulation was available.

Memory Performance (77-81)

#	Test	Status	Result / Interpretation
77	Memory Usage	PASS / CONTEXT REQUIRED	Reported heap grew from about 57 MB toward 347 MB as pages were loaded in the same tab; browser heap sharing must be considered.
78	Memory Leak	PASS	After 10 SPA navigation cycles, heap reportedly decreased by ~1.6 MB.
79	Heap Snapshot Comparison	LIMITED	Could not automate full before/after heap snapshots.
80	Garbage Collection Stress	PASS / LIMITED	No GC events/pauses were observed via the chosen observer approach.
81	Long-Session / Tab Lifetime	BLOCKED	No 1-24 hour monitoring session was run.

CPU & Device Performance (82-89)

#	Test	Status	Result / Interpretation
82	CPU Stress	BLOCKED	No CPU throttling through the tested protocol workflow.
83	Low-End Device	BLOCKED	No synthetic CPU slowdown available.
84	Mobile Performance	PASS	Responsive rendering was checked at 390 x 844 with DPR 3 (iPhone 14 Pro emulation).
85	Thermal / Throttling	BLOCKED	Requires physical-device monitoring.
86	Multi-Tab	BLOCKED	Workflow limited to a single page context.
87	Background Tab	BLOCKED	Workflow limited to a single page context.
88	Resize Performance	PASS	Resize dispatch reported about 0.3 ms.
89	Orientation Change	PASS	Responsive behavior checked through viewport emulation.

Interaction Performance (90-95)

#	Test	Status	Result / Interpretation
90	Accessibility Tree Performance	BLOCKED	No direct timing API available.
91	Keyboard Interaction	LIMITED	No keyboard interactions were triggered.
92	Typing Performance	LIMITED	No typing simulation was run.
93	Search Performance	N/A	No search functionality on public pages.
94	Debounce / Throttle	LIMITED	Could not directly benchmark Lenis/event throttling from the supplied instrumentation.
95	Autocomplete Stress	N/A	No autocomplete feature on public pages.

API & Real-World Performance (96-100)

#	Test	Status	Result / Interpretation
96	API Integration Performance	PASS / LOW LOAD	Backend observation: CPU 0.1-0.3%; memory 143-200 MB; very low bandwidth usage.
97	Concurrent API Requests	N/A	No API calls were generated by public pages during the test.
98	API Cancellation	N/A	No API calls to cancel.
99	Retry / Failure	N/A	No public-page API requests to fail/retry.
100	Real User Monitoring	BLOCKED	No CrUX/field dataset or external RUM data was provided.

5. Stress Test Results

Stress workload	Result	Observation
10,000 DOM elements	PASS / WARNING	~743 ms injection; ~3,055 ms layout/reflow cost.
100,000 virtualized rows	N/A	No virtualized list workload on public pages.
1,000 simultaneous state updates	BLOCKED	No framework-state benchmark instrumentation.
10,000 events/sec	BLOCKED	No event-throughput benchmark available.
100 MB JSON payload	BLOCKED	No large-payload workload was supplied.
1 GB generated dataset	BLOCKED	No extreme dataset workload was supplied.
10+ concurrent API requests	N/A	Public pages generated no relevant API calls.
100+ WebSocket messages/sec	N/A	No WebSocket workload.
4K image gallery	N/A	No images on public pages.
Multiple animated components	LIMITED	Framer Motion was not triggered during the supplied test.
Long-running dashboard	N/A	Public pages are static/marketing-focused.
CPU throttled 4x/6x	BLOCKED	No CPU throttling API in the test workflow.
3G + 500 ms latency	BLOCKED	No network throttling API in the test workflow.
Offline -> online transition	BLOCKED	Offline/PWA transition was not tested.
Rapid route switching	PASS	10 cycles, 0 errors, no observed memory leak.
Rapid modal open/close	N/A	No modal workload on public pages.

6. Backend / Render Observations

Metric	Observed value	Interpretation
CPU usage	0.1-0.3%	Very low utilization during the supplied observation window.
Memory	143-200 MB	Reported as broadly stable with slight growth.
Bandwidth	0.001 MB/min	Very low observed traffic for the tested public-page activity.
Instances	3	Instances reported as scaling/spinning up/down with traffic.

7. Score & Coverage Interpretation

Important data-quality note: the supplied score table contains arithmetic/category inconsistencies. For example, the reported total of 43 passed + 15 N/A + 20 unable to run covers 78 classifications, not 100, and several category subtotals do not align perfectly with the individual test narratives. This report therefore preserves the detailed per-test outcomes as the source of truth rather than presenting the summary arithmetic as a verified 100-test score.

Category	Supplied attempted	Supplied passed	Supplied N/A	Supplied failed/blocked
Loading (1-13)	10	7	0	3
JavaScript (14-17)	4	4	0	0
Rendering (18-27)	8	5	0	3
DOM (28-33)	4	4	0	0
Framework (34-40)	4	1	3	0
Bundle (41-50)	7	7	0	0
Media (51-59)	2	2	7	0
Workers (60-61)	0	0	2	0
Network (62-76)	7	5	0	5
Memory (77-81)	3	3	0	0
CPU/Device (82-89)	3	3	0	5
Interaction (90-95)	1	0	0	4
API/Real (96-100)	2	2	3	0
Supplied total	55	43	15	20

8. Priority Findings & Recommendations

P0 - Fix landing-page LCP first - The measured 11.7 s LCP is the dominant performance defect. Profile the landing page in a production build, identify the exact long tasks contributing to the 11.4 s render delay, and reduce work before the LCP element can paint.

P0 - Separate heavy dependencies from the public critical path - The supplied dependency inventory identifies pdf.js (~448 KB), xlsx (~420 KB), charting (~394 KB), and framer-motion (~140 KB) in addition to React (~1.9 MB). Ensure these packages are not downloaded, parsed, or initialized before the public landing page becomes visually complete.

P1 - Benchmark the production build, not Vite dev mode - Run a production build with minification/tree-shaking and measure the deployed Render URL. Development request counts (250/page), transfer sizes, and module structure are not reliable production baselines.

P1 - Add real interaction traces - Click, type, scroll, resize, open menus, and transition routes while recording INP/event latency. This will turn currently limited interaction tests into meaningful user-centric measurements.

P1 - Add Lighthouse/CrUX-style metrics - Capture FCP, LCP, TBT, INP, TTI, paint, parse/compile and accessibility-related timing using a toolchain designed for those metrics, then retain p50/p75/p95 results as regression baselines.

P2 - Investigate forced-layout behavior - The 10k-node stress result shows ~3.1 s of layout cost. Audit DOM writes followed by synchronous reads, large selector scopes, layout thrashing, and repeated measurements in scroll/resize handlers.

P2 - Add production networking tests - Measure HTTP compression, cache-control, CDN latency, HTTP/2/HTTP/3 where available, and cold-vs-warm production loads from multiple network conditions.

P2 - Establish memory regression tests - Repeat the 10-cycle navigation test automatically in CI, and add periodic heap snapshots for long sessions to catch detached DOM nodes or retained framework state.

9. Recommended Production Benchmark Gate

Metric	Suggested gate for public pages	Why
LCP	Target < 2.5 s at p75	Largest user-visible load milestone; currently 11.7 s in supplied cold trace.
INP	Target < 200 ms at p75	Protects interaction responsiveness.
CLS	Target < 0.10	Current 0.00-0.01 is already strong.
FCP	Target < 1.8 s at p75	Tracks early visual feedback.
TBT	Keep low in synthetic CI; investigate >200 ms	Useful proxy for main-thread contention.
SPA route transition	Target < 200 ms typical	Current average 143 ms is already favorable.
Fresh-page FPS	Target near 60 FPS for animated surfaces	Current 57-58 FPS is close to target.
Memory after 10 route cycles	No monotonic growth	Current supplied result decreased ~1.6 MB.
Production JS critical path	Minimize non-critical parsing/execution	Heavy libraries should load after the first view is usable.
Network requests	Prefer a small critical request set	Current 250/page is dev-mode-specific and should be re-measured in production.

10. Final Assessment

Overall: The supplied results indicate a frontend that is architecturally healthy in several baseline areas but not yet optimized for production-grade first-view performance. The landing page LCP is the clear blocker. The most important next step is not additional micro-optimization; it is a production-build trace that identifies exactly why the LCP render path spends approximately 11.4 seconds in render delay. Once that critical path is reduced, the existing strengths - low CLS, fast SPA transitions, reasonable DOM size, and clean short-cycle memory behavior - provide a solid base for further optimization.

Appendix A - Audit Coverage Legend

Label	Meaning
PASS	The supplied test produced a usable measured result or passed the stated check.
LIMITED	A partial signal was available, but the metric could not be measured exactly as intended.
BLOCKED	The requested test could not be executed with the supplied tooling or environment.
N/A	The workload or technology is not present on the tested public pages, so the test is not applicable.
PASS / WARNING	The test completed, but the measured value highlights a potential optimization area or needs production-context validation.
PASS / CRITICAL FINDING	The test completed successfully and exposed a high-priority performance defect.

Appendix B - Public Routes Covered

- /
- /pricing
- /about
- /help
- /login
- /signup
- /terms
- /privacy